

Factory Intelligence System

AI-Powered Worker Tracking & Geo-Mapping

MONTH 1 · 30 DAYS

Factory Understanding & Mapping Foundation

Objective

- Understand factory layout, rooms, and physical structure
- Identify camera positions and calculate field-of-view coverage
- Build base geo-mapping coordinate system using Homography
- Validate basic movement flow — entry → exit detection per camera zone

What Will Be Done

1. CCTV Coverage Analysis

Identify all camera positions (X, Y on floor plan), facing direction, and approximate coverage area using Field-of-View geometry. Map blind spots.

2. Floor Plan Digitization

Convert client-provided factory PDF into a digital coordinate system. Mark all rooms, entry/exit points, and machine zones precisely.

3. Geo-Mapping — Homography Setup

Apply homographic transformation to map real camera pixel coordinates onto the floor plan. Define anchor points at doors, corners, and key landmarks.

4. Movement Flow Validation

Track movement within a single camera feed. Define entry/exit points per zone. Validate: 'Person moves from Point A → Point B — correctly detected.'

5. Zone & Node Creation

Create key checkpoint nodes and named zones on the mapped floor plan. Example — Node A = Room Exit, Node B = Corridor Start.

Technology Stack

YOLOv8	OpenCV	Homography	NumPy	Pandas
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ByteTrack and Re-ID are intentionally excluded from Module 1 — centroid-shift logic is sufficient for movement validation in this phase.

Deliverables

- ✓ Digitized floor map with coordinate system overlay
- ✓ Camera positions and coverage zones marked on map
- ✓ Movement flow diagram (basic Spaghetti chart)
- ✓ Zone/node map with labeled checkpoints
- ✓ Technical feasibility report

Live AI Detection & Tracking System

Objective

- Deploy real-time person detection on live CCTV feeds
- Assign persistent IDs to each worker and track across zones
- Record movement distance, time-in-zone, and presence logs
- Display live results on an operational dashboard

What Will Be Built

1. Limited Live Camera Testing (1 to 3 Cameras)

We will NOT connect all cameras in this phase. Testing will be done on 1 to 3 selected cameras only. This ensures stable, accurate results before full-scale deployment.

2. Person Detection on Test Cameras

YOLOv8 runs on the selected camera feeds to detect workers in real time. Detection is validated per camera before moving to the next.

3. Worker ID Tracking

Each detected worker receives a persistent numeric ID. ByteTrack maintains IDs across frames within the selected camera feeds.

4. Movement Metrics

System records distance traveled per worker and time spent inside each defined zone per session — exported as CSV logs.

5. Presence Monitoring

Automatic logging of entry time, exit time, and total factory presence duration per worker ID per shift.

6. Demo Result Delivery

A working demo will be presented showing live detection results. If live streaming is not achievable due to network/hardware constraints, the system will process uploaded video files (MP4 format) and present results — same AI logic, same output, just pre-recorded input.

7. Live Dashboard

Streamlit dashboard showing: annotated feed, active/idle worker count, zone occupancy, and movement logs from test cameras.

Technology Stack

YOLOv8	OpenCV	ByteTrack	NumPy / Pandas	Streamlit
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Deliverables

- ✓ Live AI detection on 1 to 3 selected test cameras

- ✓ Worker ID tracking with persistent labels per session
- ✓ Movement and zone-time logs (CSV export)
- ✓ Presence report — entry / exit / duration per shift
- ✓ Demo result: live feed OR processed video upload (if live not feasible)
- ✓ Live operational dashboard (Streamlit)

What We Need From You

1. Factory Layout (PDF / CAD)	Existing factory drawing with room labels and dimensions. Cameras must be marked on the plan — even rough pencil marks are acceptable.
2. CCTV Architecture	Network/physical diagram showing camera model, IP or DVR access method, resolution, and connection type (RTSP / USB / DVR port).
3. Sample Videos — 20 clips × ~1 min	20 short video clips (~1 minute each) from all available cameras. Must include workers moving inside rooms AND between rooms.
4. Clear Room Labels	Confirmed names for every zone on the factory plan — e.g., Packing Room, Raw Material Store, Cooling Room, Reactor Area.

Sir, for ultra-accurate AI geo-mapping, a flat PDF creates scaling errors. We request the Original AutoCAD (.dwg / .dxf) file of this floor plan. Secondly, we need your IT/Security team to provide a Camera Layout Plan showing the exact IP camera placements and their viewing angles (Cone of Vision). This will allow our AI to mathematically define all blind spots accurately in Module 1.

Project Investment — Month 1

Module	Duration	Scope Summary	Cost (PKR)
Module 1	Day 1 – 15	Mapping, geo-setup, zone creation	~ 50,000
Module 2	Day 16 – 30	Live detection, tracking, dashboard	~ 60,000
TOTAL			~ 1,10,000 PKR

Payment milestone schedule to be agreed before project start. Figures are approximate and subject to final scope confirmation after client walkthrough.

This proposal covers Month 1 only. Month 2 scope (Re-ID, advanced multi-camera tracking, analytics reports) will be defined after Month 1 deliverables are reviewed and signed off.

Development Hardware Requirements

The following hardware specification is required to run the AI detection and tracking system at the performance level needed for this project:

Component	Specification
CPU	AMD Ryzen 5 5600
GPU	NVIDIA RTX 3060 (12GB / 32GB)
RAM	32 GB DDR4 ← Critical for AI workloads
Storage	1 TB NVMe SSD
PSU	650W 80+ Gold Certified
Motherboard	AMD B450 / B550
Monitor	27-inch 1440p IPS

32GB RAM is critical — required to simultaneously run YOLOv8 inference, ByteTrack, OpenCV multi-stream processing, and the Streamlit dashboard without bottleneck.

Project Agreement

This agreement is entered into between the Developer and the Client for the development of the Factory Intelligence System as described in this proposal document.

Scope: Work is limited to deliverables defined in Module 1 and Module 2 of Month 1. Any additional features or changes require a separate written agreement.

Payment: Total cost is approximately PKR 1,10,000 for Month 1. Milestone schedule to be mutually agreed before work commences.

Timeline: 30 days from agreement signing, subject to timely provision of all required materials from the Client.

Client Obligations: Client agrees to provide all required materials (factory layout, CCTV details, sample videos, room labels) within 3 working days of signing.

Confidentiality: Both parties agree to keep all project details and data confidential and not disclose to any third party without prior written consent.

Intellectual Property: Upon full payment, all deliverables become property of the Client. Developer retains the right to reference this project in their portfolio.

Developer	Client
Signature	Signature
Full Name	Full Name
Date	Date

By signing above, both parties confirm they have read, understood, and agree to all terms outlined in this proposal and agreement document.